
Calculators and communication devices are not allowed.

Answer the following questions.

1. (a) In $\mathbb{R}^4$, consider the points $P_1(-1, 2, 0, -1)$, $P_2(-1, 0, 6, -4)$ and $P_3(0, 1, 5, -3)$.
Let $X = \overrightarrow{P_1P_2}$ and $Y = \overrightarrow{P_2P_3}$.
 - (I) (5 pts.) Find a unit vector in the direction of X .
 - (II) (5 pts.) Find $d(X, Y)$, the distance between X and Y .
- (b) (10 pts.) Determine whether $V_1 = (1, 0, 0, 0)$, $V_2 = (0, 0, -2, -4)$, $V_3 = (-2, 0, 7, 14)$ and $V_4 = (0, 1, 0, 0)$ form a basis for $\mathbb{R}^4$. Justify your answer.
2. (a) Let W be a vector in $\mathbb{R}^n$, and k be any scalar.
 - (I) (5 pts.) Show that $\|kW\| = |k| \|W\|$.
 - (II) (5 pts.) Show that if $\|W\| = 0$, then $W = O$, where O is the zero vector in $\mathbb{R}^n$.
- (b) Let U and V be vectors in $\mathbb{R}^n$.
 - (I) (5 pts.) Show that if U and V are nonzero vectors with angle $\theta = 0$ between them, then $\|U + V\| = \|U\| + \|V\|$.
 - (II) (10 pts.) Show that $\|U + V\|^2 + \|U - V\|^2 = 2(\|V\|^2 + \|U\|^2)$.
3. (a) (5 pts.) Prove that the zero vector of a vector space V is unique.
- (b) (10 pts.) Prove that the negative of a vector U in a vector space V is unique.
4. Let $W = \{(a, 2a, b, a - b) : a, b \in \mathbb{R}\}$ be a subset of $\mathbb{R}^4$.
 - (a) (5 pts.) Show that W is a subspace of $\mathbb{R}^4$.
 - (b) (5 pts.) Find a basis for W .
5. (6+2+2 pts.) Let $A = \begin{bmatrix} 1 & 0 & -1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 1 \end{bmatrix}$.
 - (a) Find a basis for the null space of A .
 - (b) Find a basis for the row space of A .
 - (c) Find a basis for the column space of A .
6. In each of the following questions, select the correct answer. (2 pts. each)
 - (I) If A is an $n \times n$ matrix and U and V are $n \times 1$ matrices, then $AU \cdot V =$
 - (A) $(U \cdot V)A$
 - (B) $U \cdot AV$
 - (C) $A^T V \cdot U$
 - (D) None of the above
 - (II) The following subset of $\mathbb{R}^3$ is **not** a subspace of $\mathbb{R}^3$,
 - (A) $\{(a, 2a, b) : a, b \in \mathbb{R}\}$
 - (B) $\{(a, b, c) : 2a - b + c = 1\}$

- (III) Let $S = \{V_1, V_2, V_3, V_4\}$ be a linearly independent set of vectors in a vector space V .
- (A) If $\dim(V) = 4$, then $V = \text{span}(S)$
 - (B) If $\dim(V) > 4$, then S is a basis for V
 - (C) If $\dim(V) < 4$, then S is a basis for V
 - (D) If $W = \text{span}(S)$, then $\dim(W) < 4$
- (IV) Let $S = \{V_1, V_2, V_3\}$ be a linearly dependent set of vectors in a vector space V and $T = \{V_1 + V_2, V_2 + V_3, V_1 + V_3\}$. Then
- (A) $\dim(\text{span}(T)) = 3$
 - (B) T is linearly independent
 - (C) T is linearly dependent
 - (D) $\dim(\text{span}(S)) = 3$
- (V) Let O be the zero vector of a vector space V and $W = \{O\}$, then
- (A) W is a subspace of V and $\dim(W) = 1$
 - (B) W is a subspace of V and $\dim(W) = 0$
 - (C) W is a subspace of V and $\dim(W) = \emptyset$
 - (D) W is not a subspace of V
- (VI) Let $V = \{(a, b, c) : a > 0\}$ be a subset of $\mathbb{R}^3$. Then
- (A) V is closed under addition
 - (B) V is closed under scalar multiplication
 - (C) V is a subspace of $\mathbb{R}^3$
 - (D) None of the above
- (VII) Let V be the set of all triples of real numbers with the standard vector addition but with scalar multiplication defined by $k(x, y, z) = (k^2x, k^2y, k^2z)$. Then
- (A) V is a vector space
 - (B) If $X \in V$ and $Y \in V$, then $(X + Y) \notin V$
 - (C) V is not a vector space
 - (D) None of the above
- (VIII) Let X_1, X_2 and X_3 be any vectors in a vector space V . Let $W_1 = \text{span}\{X_1, X_2\}$ and $W_2 = \text{span}\{X_1, X_3\}$.
- (A) $\{X_1\}$ is a subspace of $W_1 \cap W_2$
 - (B) $\{X_1\} = W_1 \cap W_2$
 - (C) $W_1 \cap W_2 = \{O\}$, where O is the zero vector of V .
 - (D) None of the above
- (IX) If $S = \{X_1, X_2, \dots, X_n\}$ is a set of vectors in a vector space V and $W = \text{span}(S)$, then
- (A) $\dim(W) < n$
 - (B) $\dim(W) > n$
 - (C) $\dim(W) = n$
 - (D) None of the above
- (X) For all subspaces W_1 and W_2 of a vector space V , then
- (A) $W_1 \cup W_2$ is a subspace of V
 - (B) $W_1 \cup W_2$ is not a subspace of V
 - (C) $W_1 \cup W_2 = V$
 - (D) None of the above

1. (a) $X = \overrightarrow{P_1P_2} = (0, -2, 6, -3), Y = \overrightarrow{P_2P_3} = (1, 1, -1, 1).$

(I) $\|X\| = \sqrt{0^2 + (-2)^2 + 6^2 + (-3)^2} = 7.$

$\therefore U = \frac{1}{\|X\|}X = \frac{1}{7}(0, -2, 6, -3) = (0, -\frac{2}{7}, \frac{6}{7}, -\frac{3}{7}).$

(II) $X - Y = (-1, -3, 7, -4) \Rightarrow$

$d(X, Y) = \|X - Y\| = \sqrt{(-1)^2 + (-3)^2 + 7^2 + (-4)^2} = \sqrt{75} = 5\sqrt{3}.$

(b) $\begin{vmatrix} 1 & 0 & -2 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & -2 & 7 & 0 \\ 0 & -4 & 14 & 0 \end{vmatrix} = \begin{vmatrix} 0 & 0 & 1 \\ -2 & 7 & 0 \\ -4 & 14 & 0 \end{vmatrix} = \begin{vmatrix} -2 & 7 \\ -4 & 14 \end{vmatrix} = -28 + 28 = 0 \Rightarrow \{V_1, V_2, V_3, V_4\}$
is not a basis for $\mathbb{R}^4$.

2. (a) (I) $W = (w_1, w_2, \dots, w_n), kW = (kw_1, kw_2, \dots, kw_n) \Rightarrow$

$\|kW\| = \sqrt{(kw_1)^2 + (kw_2)^2 + \dots + (kw_n)^2} = \sqrt{k^2(w_1^2 + w_2^2 + \dots + w_n^2)}$
 $= \sqrt{k^2} \sqrt{w_1^2 + w_2^2 + \dots + w_n^2} = |k| \|W\|.$

(II) $\|W\| = 0 \Rightarrow \sqrt{w_1^2 + w_2^2 + \dots + w_n^2} = 0 \Rightarrow w_1^2 + w_2^2 + \dots + w_n^2 = 0$
 $\therefore w_1 = w_2 = \dots = w_n = 0 \Rightarrow W = (0, 0, \dots, 0) = O.$

(b) (I) $\|U + V\|^2 = (U + V) \cdot (U + V) = \|U\|^2 + 2U \cdot V + \|V\|^2$
 $= \|U\|^2 + 2\|U\|\|V\|\cos\theta + \|V\|^2 = \|U\|^2 + 2\|U\|\|V\|\cos 0 + \|V\|^2$
 $= \|U\|^2 + 2\|U\|\|V\| + \|V\|^2 = (\|U\| + \|V\|)^2 \Rightarrow \|U + V\| = \|U\| + \|V\|.$

(II) $\|U + V\|^2 + \|U - V\|^2 = (U + V) \cdot (U + V) + (U - V) \cdot (U - V)$
 $= [\|U\|^2 + 2U \cdot V + \|V\|^2] + [\|U\|^2 - 2U \cdot V + \|V\|^2] = 2(\|V\|^2 + \|U\|^2).$

3. (a) Suppose, O_1 and O_2 are two zero vectors in V . Then:

$O_1 + O_2 = O_1$ (O_2 is a zero vector) $\Rightarrow O_1 = O_2.$
 $O_1 + O_2 = O_2$ (O_1 is a zero vector)

(b) Suppose X and Y are two negatives for U . Then $\frac{U + X = O = X + U}{U + Y = O = Y + U}$. Thus,

$X = X + O = X + (U + Y) = (X + U) + Y = O + Y = Y.$

4. (a) $(a, 2a, b, a - b) = a(1, 2, 0, 1) + b(0, 0, 1, -1) \Rightarrow W = \text{span}\{(1, 2, 0, 1), (0, 0, 1, -1)\}.$
 $\therefore W$ is a subspace of $\mathbb{R}^4$.

(b) $\{(1, 2, 0, 1), (0, 0, 1, -1)\}$ is a basis for W .

5. (a) $A = \begin{bmatrix} 1 & 0 & -1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 1 \end{bmatrix} \equiv \dots \equiv \begin{bmatrix} 1 & 0 & -1 & 1 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \Rightarrow$ The solution of $AX = O$:

$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} r - s \\ -2r \\ r \\ s \end{bmatrix} = r \begin{bmatrix} 1 \\ -2 \\ 1 \\ 0 \end{bmatrix} + s \begin{bmatrix} -1 \\ 0 \\ 0 \\ 1 \end{bmatrix} \Rightarrow$

basis for the null space of $A = \left\{ \begin{bmatrix} 1 \\ -2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 0 \\ 1 \end{bmatrix} \right\}.$

(c) basis for the column space of $A = \left\{ \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix} \right\}.$

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 (C) T is linearly dependent
- (V) Let O be the zero vector of a vector space V and $W = \{O\}$, then
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- (IX) If $S = \{X_1, X_2, \dots, X_n\}$ is a set of vectors in a vector space V and $W = \text{span}(S)$, then
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